Motion in a Plane

Vectors · velocity & acceleration in 2D · relative velocity · projectile · circular motion

Class notes, Adithya Vikram

Chapter map (everything in the syllabus)

1. Scalars & vectors, types
2. Adding & subtracting vectors
3. Resolution, unit vectors, components
4. Dot & cross product
5. Position, displacement, velocity, accel^{n}
6. Constant acceleration in a plane
7. Relative velocity, rain-man
8. River-boat problems
9. Projectile motion (ground to ground)
10. Projectile from a height, on an incline
11. Uniform circular motion
12. Non-uniform circular, radius of curvature

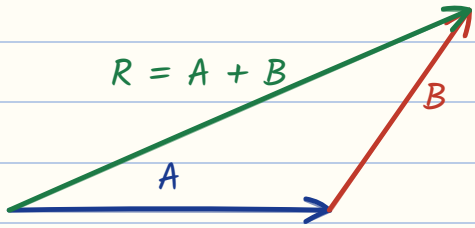
1. Scalars and vectors

- **Scalar**: only size (magnitude). e.g. mass, time, distance, speed, work, energy.
- **Vector**: size **+** direction, and adds by the **triangle law**. e.g. displacement, velocity, acceleration, force.
- Current has direction but is **NOT** a vector (it doesn't add by triangle law). Classic trap.

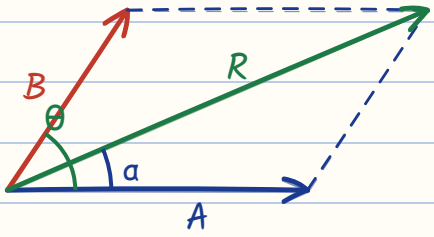
Types of vectors

- **Equal**: same magnitude, same direction (can be at different places).
- **Negative**: same magnitude, opposite direction. $\vec{A} + (-\vec{A}) = \vec{0}$.
- **Zero / null** $\vec{0}$: magnitude 0, direction not defined. e.g. displacement after a full round.
- **Unit vector** $\hat{A} = \frac{\vec{A}}{|A|}$: magnitude 1, only gives direction. $\hat{i}, \hat{j}, \hat{k}$ along x, y, z.
- **Position vector** $\vec{r}$: from origin to the point. **Displacement** $= \vec{r}_2 - \vec{r}_1$.
- **Collinear** (parallel/antiparallel) and **coplanar** (in one plane).

2. Adding vectors



triangle law: tail of B on head of A



parallelogram law

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta} \qquad \tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$$

θ = angle between A and B (tail to tail), α = angle of R from A

Special cases (learn these!)

θ	R	meaning
0°	$A + B$	maximum
90°	$\sqrt{A^2 + B^2}$	perpendicular
180°	$\lvert A - B\rvert$	minimum

$$\lvert A - B\rvert \le R \le A + B$$

any third vector outside this range can never be the resultant

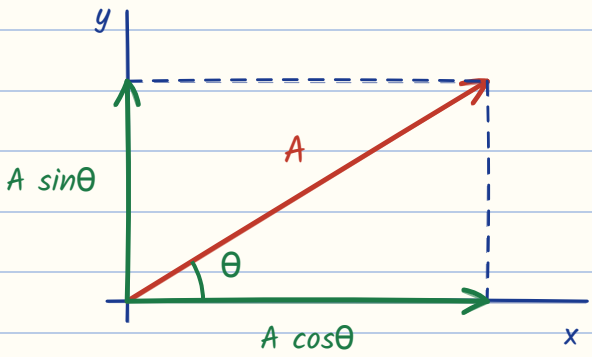
Shortcut Two **equal** vectors A at angle θ: $R = 2A\cos(\theta/2)$, and R bisects the angle. $R = A$ when $\theta = 120°$; $R = A\sqrt{2}$ at 90°; $R = A\sqrt{3}$ at 60°.

Subtraction

$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$. $\lvert\vec{A} - \vec{B}\rvert = \sqrt{A^2 + B^2 - 2AB\cos\theta}$

- Addition is commutative and associative; subtraction is **not** commutative.
- Polygon law: many vectors head-to-tail; closing side (start → end) is the resultant. Closed polygon ⇒ resultant 0.

3. Resolution and components



Split $\vec{A}$ along x and y:

$$A_x = A\cos\theta, \quad A_y = A\sin\theta$$

$$A = \sqrt{A_x^2 + A_y^2}, \quad \tan\theta = \frac{A_y}{A_x}$$

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k}, \quad |A| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

Direction cosines: $\cos\alpha = \frac{A_x}{A}$, $\cos\beta = \frac{A_y}{A}$, $\cos\gamma = \frac{A_z}{A}$ $\Rightarrow$ $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Adding by components (the fastest way)

$\vec{A} + \vec{B} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j} + (A_z + B_z)\hat{k}$. Add i with i, j with j, k with k.

★ **IMP** A vector can be resolved along **any** two directions, but components along **perpendicular** axes are the useful ones. Component along a direction = A cos(angle with that direction).

Multiplying by a number

- $\lambda\vec{A}$: magnitude $|\lambda|A$; same direction if $\lambda > 0$, opposite if $\lambda < 0$.
- Multiplying by a scalar with units (e.g. mass × velocity) gives a new physical quantity (momentum).

Shortcut To find a vector of magnitude k along $\vec{A}$: $k\,\hat{A} = k\,\frac{\vec{A}}{|A|}$.

e.g. A 10 N force at 37° to the x-axis: $F_x = 10\cos37° = 10 \times \frac{4}{5} = 8$ N, $F_y = 10 \times \frac{3}{5} = 6$ N.

e.g. $\vec{A} = 3\hat{i} + 4\hat{j}$: $|A| = \sqrt{9 + 16} = 5$, $\hat{A} = \frac{3\hat{i} + 4\hat{j}}{5}$, angle with x = $\tan^{-1}\frac{4}{3} \approx 53.1°$ (the 37–53 triangle!).

4. Dot (scalar) product

$$\vec{A}\cdot\vec{B} = AB\cos\theta = A_xB_x + A_yB_y + A_zB_z$$

answer is a scalar

- $\hat{i}\cdot\hat{i} = \hat{j}\cdot\hat{j} = \hat{k}\cdot\hat{k} = 1$, $\hat{i}\cdot\hat{j} = 0$ ...
- Angle between: $\cos\theta = \frac{\vec{A}\cdot\vec{B}}{AB}$. $\vec{A} \perp \vec{B} \Leftrightarrow \vec{A}\cdot\vec{B} = 0$.
- Projection of $\vec{A}$ on $\vec{B} = \frac{\vec{A}\cdot\vec{B}}{B}$. Work $W = \vec{F}\cdot\vec{s}$.
- Commutative: $\vec{A}\cdot\vec{B} = \vec{B}\cdot\vec{A}$.

Cross (vector) product

$$|\vec{A}\times\vec{B}| = AB\sin\theta, \quad \text{direction} \perp \text{to both (right-hand rule)}$$

$$\vec{A}\times\vec{B} = (A_yB_z - A_zB_y)\hat{i} - (A_xB_z - A_zB_x)\hat{j} + (A_xB_y - A_yB_x)\hat{k}$$

write as a 3×3 determinant with i, j, k on top

- $\hat{i}\times\hat{j} = \hat{k}$, $\hat{j}\times\hat{k} = \hat{i}$, $\hat{k}\times\hat{i} = \hat{j}$ (cyclic); reverse order ⇒ minus sign. $\hat{i}\times\hat{i} = 0$.
- **Not** commutative: $\vec{A}\times\vec{B} = -\vec{B}\times\vec{A}$. Parallel vectors ⇒ cross = 0.
- $|\vec{A}\times\vec{B}|$ = area of the parallelogram; half of it = area of the triangle.

Shortcut Check your cross product: dot it with $\vec{A}$ and with $\vec{B}$. Both must be 0. Also $|A\times B|^2 + (A\cdot B)^2 = A^2B^2$.

Careful! NCERT keeps dot product in Work–Energy and cross product in Rotation, but JEE asks both with vectors, so learn them here.

e.g. Find λ so that $(2\hat{i} + \lambda\hat{j}) \perp (3\hat{i} - 6\hat{j})$: dot $= 6 - 6\lambda = 0 \Rightarrow \lambda = 1$.

e.g. Area of the triangle formed by $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$ and $\vec{B} = \hat{i} - 2\hat{j} + 2\hat{k}$: $\frac{1}{2}|\vec{A}\times\vec{B}| = \frac{1}{2}\sqrt{90} \approx 4.74$ unit².

Shortcut Unit vector $\perp$ to both $\vec{A}$ and $\vec{B}$: $\hat{n} = \pm\frac{\vec{A}\times\vec{B}}{|\vec{A}\times\vec{B}|}$.

5. Position, velocity, acceleration in a plane

$$\vec{r} = x\hat{i} + y\hat{j}, \quad \Delta\vec{r} = \Delta x\hat{i} + \Delta y\hat{j}$$

- Average velocity $\vec{v}_{avg} = \frac{\Delta\vec{r}}{\Delta t}$ (direction of displacement).
- Instantaneous velocity $\vec{v} = \frac{d\vec{r}}{dt} = v_x\hat{i} + v_y\hat{j}$; $v_x = \frac{dx}{dt}$, $v_y = \frac{dy}{dt}$.
- Velocity is always along the tangent to the path. Its angle: $\tan\theta = \frac{v_y}{v_x}$.
- Acceleration $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$. It can point anywhere, not only along $\vec{v}$.
- Speed $= |\vec{v}| = \sqrt{v_x^2 + v_y^2}$. Average speed (distance/time) $\geq$ |average velocity|.

★ **IMP** $a \parallel v \Rightarrow$ speed changes, path straight. $a \perp v \Rightarrow$ only direction changes (speed constant). In general a has both parts.

6. Constant acceleration in a plane

$$\vec{v} = \vec{u} + \vec{a}t, \quad \vec{r} = \vec{r}_0 + \vec{u}t + \tfrac{1}{2}\vec{a}t^2$$

The **x and y motions are independent**. Use the 1-D equations separately:

$$x = u_x t + \tfrac{1}{2}a_x t^2, \quad y = u_y t + \tfrac{1}{2}a_y t^2, \quad v_x^2 = u_x^2 + 2a_x x \ \ldots$$

Shortcut Given r(t) or x(t), y(t): differentiate once for v, twice for a. Eliminate t to get the path y(x).

e.g. $x = 3t$, $y = 4t - 5t^2$ (SI). Then $v_x = 3$, $v_y = 4 - 10t$, so at $t = 0$ the speed is 5 m/s at 53.1°; $a_x = 0$, $a_y = -10$ m/s². Put $t = x/3$: $y = \frac{4}{3}x - \frac{5}{9}x^2$, a parabola. It's just a projectile!

e.g. $\vec{u} = 2\hat{i}$ m/s, $\vec{a} = 2\hat{j}$ m/s², $t = 3$ s: $\vec{v} = 2\hat{i} + 6\hat{j}$ m/s, $\vec{r} = 6\hat{i} + 9\hat{j}$ m (start at origin).

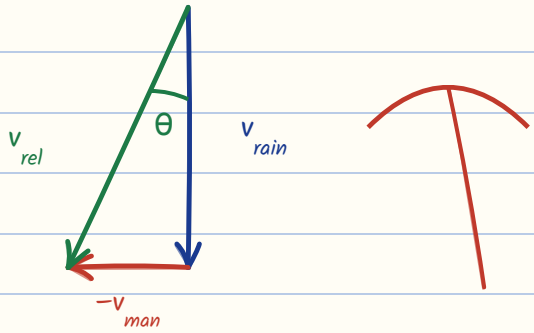
7. Relative velocity

$$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$$

velocity of A as seen by B

- $\vec{v}_{BA} = -\vec{v}_{AB}$.
- Same direction: $|v_{AB}| = v_A - v_B$. Opposite: $v_A + v_B$. At angle θ: $\sqrt{v_A^2 + v_B^2 - 2v_A v_B \cos\theta}$.

Rain–man problems



tilt the umbrella forward by θ

Rain falls vertically at v_r. You walk at v_m.

Rain relative to you:

$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$$

$$\tan\theta = \frac{v_m}{v_r}$$

θ from the vertical, tilt umbrella forward

Rain seems to hit you at $\sqrt{v_r^2 + v_m^2}$.

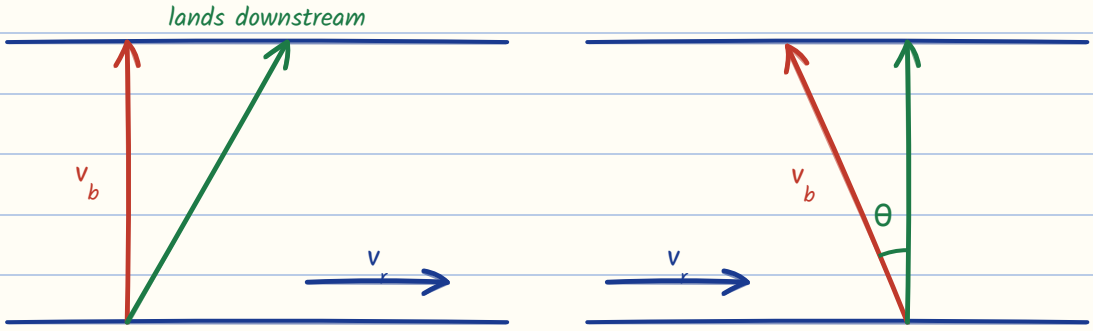
Shortcut "Rain appears vertical" means the horizontal parts cancel: $v_{rain,x} = v_{man}$. Draw the velocity triangle, then just read off the angle.

Careful! Relative velocity is about what one observer **sees**. Always write $v_{AB} = v_A - v_B$ with signs/directions first, then put numbers.

e.g. Car A goes east at 20 m/s, car B goes north at 15 m/s. $\vec{v}_{AB} = 20\hat{i} - 15\hat{j}$: $|v_{AB}| = \sqrt{400 + 225} = 25$ m/s, at 36.9° south of east. (B sees A moving away "south-east".)

Shortcut Closest approach of two moving bodies: go to the frame of one of them. The other then moves in a straight line with v_{rel}; the least distance is the perpendicular from the first body to that line.

8. River–boat problems



① shortest TIME: head straight across ② shortest PATH: aim upstream

River width d, flow v_r, boat speed in still water v_b, heading at θ with the straight-across line (upstream +).

$$\text{time } t = \frac{d}{v_b \cos\theta}, \quad \text{drift } x = (v_r - v_b \sin\theta)\cdot t$$

① **Shortest time**: head straight across ($\theta = 0$).

$$t_{min} = \frac{d}{v_b}, \quad \text{drift} = \frac{v_r\, d}{v_b}$$

② **Shortest path** (zero drift): aim upstream.

$$\sin\theta = \frac{v_r}{v_b}$$

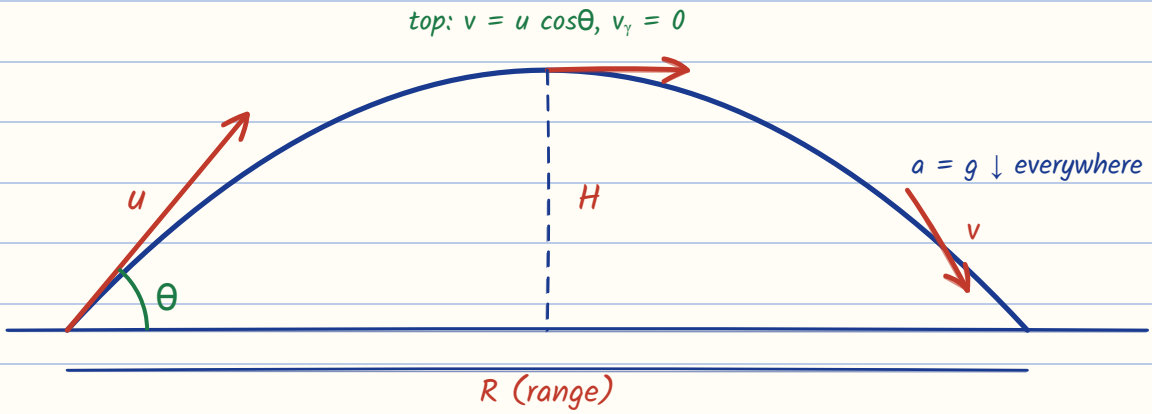
$$t = \frac{d}{\sqrt{v_b^2 - v_r^2}}$$

★ **IMP** Zero drift is possible only if $v_b > v_r$. If $v_b < v_r$, the drift is least when $\sin\theta = \frac{v_b}{v_r}$ (θ measured from the straight-across line, towards upstream).

Shortcut Shortest time never depends on the river speed: the river only carries you sideways. Aim straight, multiply time by v_r for the drift.

Same idea for an aeroplane in wind: velocity relative to ground = velocity relative to air + wind velocity.

9. Projectile motion (ground to ground)



Only gravity acts (no air resistance). x: uniform motion, y: uniformly accelerated (−g). $u_x = u\cos\theta$, $u_y = u\sin\theta$.

$$x = (u\cos\theta)t, \quad y = (u\sin\theta)t - \tfrac{1}{2}gt^2$$

Trajectory: $y = x\tan\theta - \dfrac{g x^2}{2u^2\cos^2\theta} = x\tan\theta\left(1 - \dfrac{x}{R}\right)$

a parabola

Time of flight	Max height	Range
$T = \dfrac{2u\sin\theta}{g}$	$H = \dfrac{u^2\sin^2\theta}{2g}$	$R = \dfrac{u^2\sin 2\theta}{g}$

- Velocity at time t: $v_x = u\cos\theta$ (constant), $v_y = u\sin\theta - gt$; $v = \sqrt{v_x^2 + v_y^2}$, $\tan\varphi = \dfrac{v_y}{v_x}$.
- At the top: $v_y = 0$, $v = u\cos\theta$, $\vec{v} \perp \vec{g}$. Time to top = T/2.
- Lands with the same speed u at angle θ below the horizontal.

e.g. Path $y = x - \dfrac{x^2}{20}$ (SI, g = 10). Compare with $y = x\tan\theta - \dfrac{gx^2}{2u^2\cos^2\theta}$: $\tan\theta = 1 \Rightarrow \theta = 45°$; $\dfrac{10}{2u^2(\frac{1}{2})} = \dfrac{1}{20} \Rightarrow u = \sqrt{200} \approx 14.1$ m/s. Range (y = 0): R = 20 m, H = R/4 = 5 m.

Projectile: key results & shortcuts

- **Max range** at $\theta = 45°$: $R_{max} = \frac{u^2}{g}$; then $H = \frac{R_{max}}{4}$. e.g. $u = 20$ m/s: $R_{max} = 40$ m, $H = 10$ m.
- **Complementary angles** θ and $90° - \theta$ give the **same range** (e.g. 15° & 75° at 20 m/s: 20 m each).
- For those two: $H_1 + H_2 = \frac{u^2}{2g}$, $R = 4\sqrt{H_1 H_2}$, $T_1 T_2 = \frac{2R}{g}$.
- **H and R**: $\frac{H}{R} = \frac{\tan\theta}{4} \Rightarrow \tan\theta = \frac{4H}{R}$. $H = R \Rightarrow \theta = \tan^{-1}4$. Also $\frac{H}{T^2} = \frac{g}{8}$.
- **Reaches height h twice** at t_1, t_2: $t_1 + t_2 = T$, $t_1 t_2 = \frac{2h}{g}$.
- KE at the top = $K\cos^2\theta$ (K = initial KE). Momentum change from launch to landing = $2mu\sin\theta$ (downward).
- **Radius of curvature** at the top: $r = \frac{(u\cos\theta)^2}{g}$; at launch: $r = \frac{u^2}{g\cos\theta}$.
- Angle between $\vec{v}$ and $\vec{a}$: $90° + \theta$ at launch, 90° at top, $90° - \theta$ at landing.

> **Shortcut** For any "same u, two angles, same range" question: the angles add to 90°. Write $R = 4\sqrt{(H_1 H_2)}$ or $T_1 T_2 = 2R/g$ and you're done in one line.

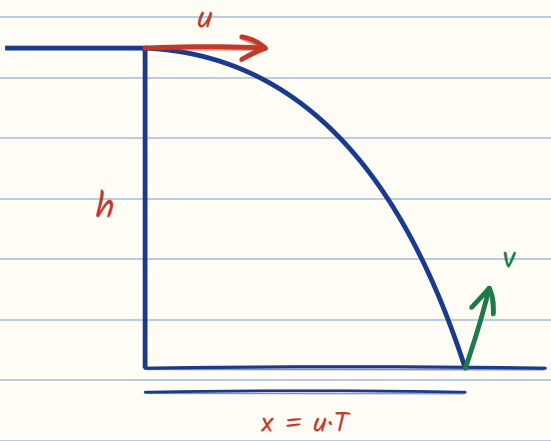
> **Careful!** These shortcuts need **same launch and landing level** and no air resistance. From a height or on an incline they change.

Must-know values

θ	0°	30°	37°	45°	53°	60°	90°
sinθ	0	½	3/5	1/√2	4/5	√3/2	1
cosθ	1	√3/2	4/5	1/√2	3/5	½	0
tanθ	0	1/√3	3/4	1	4/3	√3	∞

37°/53° are the JEE convention (from the 3-4-5 triangle); $g = 10$ m/s² unless told otherwise.

10. Projectile from a height



Thrown horizontally with speed u from height h:

$$T = \sqrt{2h/g}, \quad x = u\sqrt{2h/g}$$

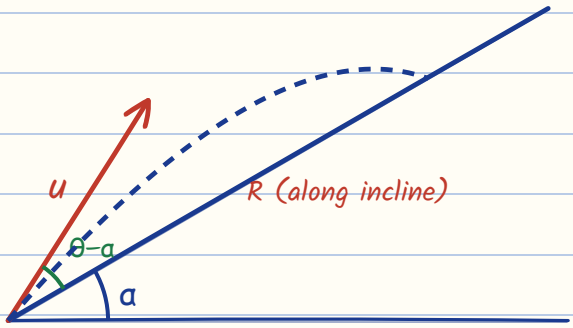
$$v_y = gT = \sqrt{2gh}, \quad v = \sqrt{u^2 + 2gh}$$

Path: $y = \frac{g x^2}{2u^2}$ (measured down from the top).

Thrown at θ from height h: use $-h = (u \sin\theta)t - \frac{1}{2}gt^2$, solve the quadratic for t (take the + root), then $x = (u \cos\theta)t$.

★ **IMP** Time of fall depends **only on h**, not on u. A ball dropped and a ball thrown horizontally from the same height land together.

Projectile on an inclined plane



Incline angle α, launched at θ to the **horizontal** (up the incline). Take axes along and ⊥ to the incline: g has parts $g \sin\alpha$ (along) and $g \cos\alpha$ (⊥).

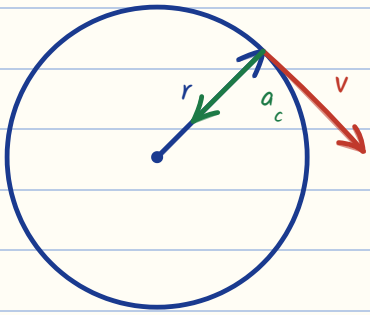
$$T = \frac{2u \sin(\theta - \alpha)}{g \cos\alpha}$$

$$R = \frac{2u^2 \sin(\theta - \alpha) \cos\theta}{g \cos^2\alpha}$$

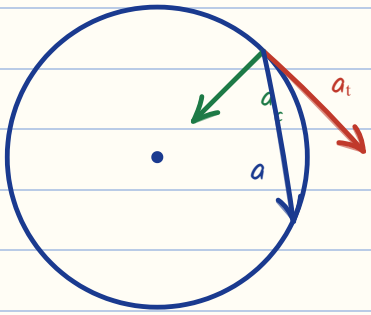
$$R_{max,\,up} = \frac{u^2}{g(1 + \sin\alpha)} \text{ at } \theta = 45° + \alpha/2; \quad R_{max,\,down} = \frac{u^2}{g(1 - \sin\alpha)}$$

Shortcut Hits the incline at right angles ⇔ the velocity along the incline is zero at landing: $u \cos(\theta - \alpha) = g \sin\alpha \cdot T$. Use the incline axes and it's 1-D.

11. Uniform circular motion



uniform: $|v|$ const, $a \perp v$



non-uniform: $a = \sqrt{(a_c^2 + a_t^2)}$

- Angular displacement $\Delta\theta$ (rad), angular velocity $\omega = \frac{d\theta}{dt}$, $\omega = \frac{2\pi}{T} = 2\pi f$.
- Speed $v = \omega r$ (constant), but the velocity **changes direction** ⇒ there is an acceleration.

$$a_c = \frac{v^2}{r} = \omega^2 r = \frac{4\pi^2 r}{T^2} \quad \text{(towards the centre)}$$

★ **IMP** In uniform circular motion speed is constant, but velocity and acceleration are **not** constant vectors (their directions keep turning). $|a|$ stays the same.

- $\vec{v} = \vec{\omega} \times \vec{r}$; $\vec{v} \perp \vec{r}$, $\vec{a} \parallel -\vec{r}$.
- In one full turn: displacement = 0, average velocity = 0, distance = $2\pi r$.
- Over half a turn: $|\Delta\vec{v}| = 2v$; over angle θ: $|\Delta\vec{v}| = 2v \sin(\theta/2)$.

12. Non-uniform circular motion

$$a_t = \frac{dv}{dt} = \alpha r \text{ (along v)}, \quad a_c = \frac{v^2}{r}, \quad a = \sqrt{a_c^2 + a_t^2}$$

angle of a with v: $\tan\varphi = a_c / a_t$

Radius of curvature of any path: $r = \frac{v^2}{a_\perp}$ ($a_\perp$ = part of a perpendicular to v). For $y(x)$: $r = \frac{(1 + y'^2)^{3/2}}{|y''|}$.

Shortcut Projectile at the top: $a_\perp = g$, $v = u\cos\theta \Rightarrow r = u^2\cos^2\theta / g$. Same rule works at any point of any curve.

One-page formula sheet

$R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$	$\tan\alpha = B\sin\theta / (A + B\cos\theta)$
equal vectors: $R = 2A\cos(\theta/2)$	$\lvert A - B\rvert \le R \le A + B$
$A_x = A\cos\theta$, $A_y = A\sin\theta$	$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
$A\cdot B = AB\cos\theta = \Sigma A_i B_i$	$\lvert A \times B\rvert = AB\sin\theta$ (area)
$v = dr/dt$, $a = dv/dt$	$v = u + at$, $r = r_0 + ut + \frac{1}{2}at^2$
$v_{AB} = v_A - v_B$	rain: $\tan\theta = v_m / v_r$
river: $t_{min} = d / v_b$	drift (shortest time) $= v_r\, d / v_b$
zero drift: $\sin\theta = v_r / v_b$	its time: $t = d / \sqrt{(v_b^2 - v_r^2)}$
$v_b < v_r$: least drift at $\sin\theta = v_b / v_r$	closest approach: use the relative frame
$T = 2u\sin\theta / g$	$H = u^2\sin^2\theta / 2g$
$R = u^2\sin 2\theta / g$	$R_{max} = u^2/g$ at 45°, $H = R/4$
$y = x\tan\theta\,(1 - x/R)$	$\tan\theta = 4H / R$
θ, $90°-\theta$: $R = 4\sqrt{(H_1H_2)}$	$T_1T_2 = 2R / g$
horizontal: $T = \sqrt{(2h/g)}$	$v = \sqrt{(u^2 + 2gh)}$
incline: $T = 2u\sin(\theta-\alpha) / g\cos\alpha$	R_{max} up $= u^2 / g(1 + \sin\alpha)$
$\omega = 2\pi/T = 2\pi f$, $v = \omega r$	$a_c = v^2/r = \omega^2 r$
$a = \sqrt{(a_c^2 + a_t^2)}$	$r_{curv} = v^2 / a_\perp$; top: $u^2\cos^2\theta / g$

Common mistakes

- Using θ = angle between **head and tail** instead of tail-to-tail in $R = \sqrt{(A^2 + B^2 + 2AB\cos\theta)}$.
- Forgetting that range formulas need launch and landing at the **same level**.
- Taking velocity at the top as zero. Only $v_y = 0$; $v = u\cos\theta$.
- Writing $v_{AB} = v_B - v_A$ (wrong order).
- Saying "uniform circular motion has zero acceleration". It has v^2/r towards the centre.

PYQ-pattern questions with solutions

Original questions written in the style of recurring JEE Main / Advanced and NCERT themes (not copied from papers). Every answer is computed and checked by code.

Q1. JEE Main pattern · vectors Two vectors of magnitude 3 and 5 make 60° with each other. Find the resultant and its angle with the first vector.

Soln:

1. $R = \sqrt{(9 + 25 + 2\cdot3\cdot5\cdot\cos60°)} = \sqrt{49} = 7$
2. $\tan\alpha = 5\sin60° / (3 + 5\cos60°) = 4.33 / 5.5 \Rightarrow \alpha \approx 38.2°$

$\Rightarrow$ **$R = 7$, $\alpha \approx 38.2°$**

Q2. JEE Main pattern · equal vectors Two forces of equal magnitude F have a resultant of magnitude F. Find the angle between them.

Soln:

1. Equal vectors: $R = 2F\cos(\theta/2) = F$
2. $\cos(\theta/2) = \tfrac{1}{2} \Rightarrow \theta/2 = 60°$

$\Rightarrow$ **$\theta = 120°$**

Q3. JEE Main pattern · dot & cross $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$, $\vec{B} = \hat{i} - 2\hat{j} + 2\hat{k}$. Find $\vec{A}\cdot\vec{B}$, the angle between them, and $\vec{A}\times\vec{B}$.

Soln:

1. $A\cdot B = 2 - 6 - 2 = -6$; $|A| = \sqrt{14}$, $|B| = 3$
2. $\cos\theta = -6 / (3\sqrt{14}) \Rightarrow \theta \approx 122°$
3. $A\times B = (3\cdot2 - (-1)(-2))\hat{i} - (2\cdot2 - (-1)(1))\hat{j} + (2(-2) - 3\cdot1)\hat{k} = 4\hat{i} - 5\hat{j} - 7\hat{k}$
4. Check: $(A\times B)\cdot A = 8 - 15 + 7 = 0$ ✓

$\Rightarrow$ **$A\cdot B = -6$, $\theta \approx 122°$, $|A\times B| = \sqrt{90} \approx 9.49$**

Q4. JEE Main pattern · calculus A particle moves with $\vec{r} = 2t^2\hat{i} + (3t - t^2)\hat{j}$ m. Find its speed at $t = 1$ s and its acceleration.

Soln:

1. $\vec{v} = 4t\hat{i} + (3 - 2t)\hat{j} \Rightarrow$ at $t = 1$: $4\hat{i} + \hat{j}$
2. speed $= \sqrt{17} \approx 4.12$ m/s
3. $\vec{a} = 4\hat{i} - 2\hat{j}$ (constant), $|a| = \sqrt{20}$

$\Rightarrow$ **4.12 m/s; $a = 4\hat{i} - 2\hat{j}$, $|a| \approx 4.47$ m/s²**

Q5. JEE Main pattern · projectile A ball is thrown at 20 m/s at 30° above the horizontal ($g = 10$). Find T, H and R.

Soln:

1. $u_x = 20\cos30° = 17.3$, $u_y = 10$ m/s
2. $T = 2u_y/g = 2$ s; $H = u_y^2/2g = 5$ m
3. $R = u^2 \sin60° / g = 400(0.866)/10 = 34.6$ m

⇒ **$T = 2$ s, $H = 5$ m, $R = 34.6$ m**

Q6. JEE Main pattern · complementary angles Two balls are thrown with the same speed at 30° and 60° ($u = 20$ m/s). Their heights are H_1 and H_2. Show the range is $4\sqrt{(H_1H_2)}$.

Soln:

1. $H_1 = 5$ m, $H_2 = 15$ m
2. $4\sqrt{(H_1H_2)} = 4\sqrt{75} = 34.6$ m $= R$ for both ✓
3. Also $H_1 + H_2 = 20$ m $= u^2/2g$ ✓

⇒ **$R = 34.6$ m**

Q7. JEE Main pattern · horizontal projection A ball is thrown horizontally at 10 m/s from a 45 m high cliff. Find the time, the distance from the foot, and the velocity on landing.

Soln:

1. $T = \sqrt{(2h/g)} = \sqrt{9} = 3$ s; $x = uT = 30$ m
2. $v_y = gT = 30$ m/s, $v = \sqrt{(100 + 900)} \approx 31.6$ m/s
3. angle below horizontal: $\tan^{-1}(30/10) \approx 71.6°$

⇒ **3 s, 30 m, 31.6 m/s at 71.6° below horizontal**

Q8. JEE Main pattern · H vs R At what angle must a projectile be thrown so that its maximum height equals its range?

Soln:

1. $H/R = \tan\theta/4 = 1$
2. $\tan\theta = 4$

⇒ **$\theta = \tan^{-1}4 \approx 76.0°$**

Q9. JEE Main pattern · river-boat A river is 100 m wide and flows at 3 m/s. A boat moves at 5 m/s in still water. (a) Least time to cross and the drift? (b) Direction and time to reach the point straight opposite?

Soln:

1. (a) head straight across: $t = 100/5 = 20$ s; drift $= 3 \times 20 = 60$ m
2. (b) $\sin\theta = 3/5 \Rightarrow \theta \approx 36.9°$ upstream from straight-across
3. speed across $= \sqrt{(25 - 9)} = 4$ m/s $\Rightarrow t = 100/4 = 25$ s

$\Rightarrow$ **(a) 20 s, 60 m; (b) 36.9° upstream, 25 s**

Q10. JEE Main pattern · rain-man Rain falls vertically at 10 m/s. A man walks at 5 m/s. At what angle should he hold his umbrella?

Soln:

1. $v_{rain,man} = v_{rain} - v_{man}$
2. $\tan\theta = 5/10 \Rightarrow \theta \approx 26.6°$ from vertical, tilted forward
3. rain seen at $\sqrt{125} \approx 11.2$ m/s

$\Rightarrow$ **26.6° forward from vertical**

Q11. NCERT / JEE pattern · uniform circular A stone tied to a 0.5 m string is whirled at 2 revolutions per second. Find ω, v and the centripetal acceleration.

Soln:

1. $\omega = 2\pi f = 4\pi \approx 12.6$ rad/s
2. $v = \omega r = 2\pi \approx 6.28$ m/s
3. $a = \omega^2 r = 8\pi^2 \approx 79.0$ m/s² (towards the centre)

$\Rightarrow$ **$\omega \approx 12.6$ rad/s, $v \approx 6.28$ m/s, $a \approx 79.0$ m/s²**

Q12. JEE Main pattern · non-uniform circular A particle on a circle of radius 2 m has speed 4 m/s, increasing at 3 m/s². Find its total acceleration.

Soln:

1. $a_c = v^2/r = 16/2 = 8$ m/s²; $a_t = 3$ m/s²
2. $a = \sqrt{(64 + 9)} = \sqrt{73} \approx 8.54$ m/s²
3. angle with v: $\tan^{-1}(8/3) \approx 69.4°$

$\Rightarrow$ **8.54 m/s² at 69.4° to the velocity**

Q13. JEE Advanced pattern · incline A ball is thrown at 20 m/s at 75° to the horizontal, up an incline of 30°. Find the time of flight and the range along the incline. What angle would give the maximum range?

Soln:

1. $T = 2u\sin(\theta - \alpha)/(g\cos\alpha) = 2\cdot20\cdot\sin45°/(10\cos30°) \approx 3.27$ s
2. $R = 2u^2\sin(\theta-\alpha)\cos\theta/(g\cos^2\alpha) = 2\cdot400\cdot(0.707)(0.259)/(10\cdot0.75) \approx 19.5$ m
3. max range at $\theta = 45° + \alpha/2 = 60°$: $R_{max} = u^2/g(1+\sin\alpha) \approx 26.7$ m

⇒ **$T \approx 3.27$ s, $R \approx 19.5$ m; best angle 60°**

Q14. JEE Main pattern · speed at top The speed of a projectile at its highest point is half its launch speed. Find the launch angle and the ratio of KE at the top to the initial KE.

Soln:

1. $u\cos\theta = u/2 \Rightarrow \cos\theta = ½$
2. $KE_{top} / KE = \cos^2\theta = ¼$

⇒ **$\theta = 60°$, ratio 1 : 4**

Q15. JEE Advanced pattern · radius of curvature Find the radius of curvature of the path at the highest point for $u = 20$ m/s at 60°.

Soln:

1. At the top $a_\perp = g$ and $v = u\cos60° = 10$ m/s
2. $r = v^2/g = 100/10$

⇒ **$r = 10$ m**

Last-minute checklist

- Resolve everything into x and y first. Most 2-D problems become two 1-D problems.
- Projectile: horizontal velocity never changes; vertical is just free fall.
- Same level? Use T, H, R formulas. Different level / incline? Use components or incline axes.
- Relative velocity: draw the triangle $v_{AB} = v_A - v_B$ before any numbers.
- Circle: $a_c = v^2/r$ always; add a_t if the speed changes.

— end of chapter: Motion in a Plane —